

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: RepoKids

Project / Product Name: TraceLens - Code Retrieval Intelligence

Organization / Institution (if any): SRM Institute of Science and Technology, Kattankulathur

Submission Date: 04/10/2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

Yes

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: NA

Code generation or assistance: Generating initial Streamlit dashboard UI components, Reciprocal Rank Fusion (RRF) scoring logic, and regex boilerplate for code normalization.

UI / UX design: NA

Content creation: NA

Data analysis: Structuring metrics comparison format for MTEB benchmarking results.

Testing / debugging: Structuring metrics comparison format for MTEB benchmarking results.

Other: NA

4. Feature Origin Classification

1. Feature 1

- a. Feature Name: Hybrid Retrieval Pipeline (FAISS + BM25 with Reciprocal Rank Fusion)
- b. Self-Generated / AI-Generated / Both: Both
- c. Description:
- d. AI Tools / Platform Used: Gemini

- e. Prompt Used: "Help implement reciprocal rank fusion formula combining FAISS dense scores and rank-bm25 scores with an optional cross-encoder reranking pass."
- f. Output Summary: A template Python function that calculates RRF fused scores $\frac{1}{60 + \text{rank}}$ and calls `sentence_transformers.CrossEncoder`.
- g. Modifications: Integrated with MTEB's AbsEncoder interface, tuned the RRF $k=60$ parameter, and added top-k candidate thresholding.

2. Feature 2

- a. Feature Name: Code Preprocessing & Algorithmic Term Extraction (`clean.py`)
- b. Self-Generated / AI-Generated / Both: Both
- c. Description:
- d. AI Tools / Platform Used: Gemini / ChatGPT
- e. Prompt Used: "Provide regex patterns to remove competitive programming input/output boilerplate and extract hints like `heapq`, `bisect`, and `dfs`."
- f. Output Summary: Basic regular expressions for stripping I/O blocks and docstrings.
- g. Modifications: Refined regex rules to prevent accidental stripping of critical logic and added specialized competitive programming keywords (`bitmask`, `modulo`, `dp`, `shortest path`).

3. Feature 3

- a. Feature Name: Incremental Indexer with SHA-256 Delta Caching (`incremental.py`)
- b. Self-Generated / AI-Generated / Both: Self-Generated
- c. Description:
- d. AI Tools / Platform Used: None (Architected and coded directly by team)
- e. Prompt Used: N/A
- f. Output Summary: Core caching logic utilizing SHA-256 hashing to track file modifications and avoid redundant full re-indexing over the free tier.
- g. Modifications: Fully custom implementation and integration with `.tracelens_cache.json`.

4. Feature 4

- a. Feature Name: Interactive Streamlit Demo App (`demo.py`)
- b. Self-Generated / AI-Generated / Both: Both
- c. Description:
- d. AI Tools / Platform Used: Gemini
- e. Prompt Used: "Create a modern dark-mode Streamlit dashboard with custom CSS, sidebar dataset uploader, latency indicator, and retrieval score breakdown."

- f. Output Summary: Streamlit frontend skeleton with custom cards and file uploader widgets.
- g. Modifications: Added real-time metric visualization (Dense Cosine, BM25, and RRF bars), Rocchio PRF toggle, and interactive cache invalidation buttons.

5. Feature 5

- a. Feature Name: Evaluation Dataset & Test Corpus Generation (test_corpus.json)
- b. Self-Generated / AI-Generated / Both: Both
- c. Description:
- d. AI Tools / Platform Used: Gemini
- e. Prompt Used: "Create a JSON file containing 22 diverse competitive programming algorithms in Python formatted as {doc_id: {text: code}} for retrieval testing."
- f. Output Summary: Structured JSON containing code implementations for Bellman-Ford, Trie, Segment Tree, Knapsack, and Two Pointers.
- g. Modifications: Verified all code snippets for syntax accuracy, added realistic function docstrings, and formulated corresponding benchmark queries.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

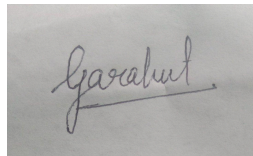
No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Garv Rahut

Role: Team Lead

Signature:

A rectangular box containing a handwritten signature in dark ink. The signature appears to be 'Garv Rahut' written in a cursive, slightly stylized script.

Date: 04/10/2026